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Electronic Component Including Mirror and Electronic Circuit

Abstract

An electronic component includes a semiconductor substrate having a first major surface and a second major surface opposing the first major surface. The electronic component further includes an electronic circuit integrated into the semiconductor substrate and a mirror arranged over the semiconductor substrate, at least a portion of the mirror being inclined with respect to the first major surface of the semiconductor substrate. The electronic component further includes at least one electrical contact configured to provide an electrical coupling between the electronic circuit and at least one electrode of an ion trap.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to electronic components including a mirror and an electronic circuit. In addition, the present disclosure relates to devices for controlling trapped ions including such electronic components.

BACKGROUND

[0002] Ion trap devices may be used for various applications, such as trapped-ion quantum computers, where dozens of trapped ions may need to be controlled with a low two-qubit error rate. Other applications may be atomic clocks. Typically, ion trap devices may require multiple lasers per (ion) processing zone and a plurality of electrodes per trapped ion. When scaling ion trap devices to larger sizes, an increasing number of the mentioned components may become a major challenge in terms of optical and electrical access to the ion trap.

[0003] It may be desirable to improve an optical and electrical access to the ion traps. In addition, the solutions developed may support a straightforward and easy scaling of ion trap devices that may increase the number of trapped ions.

SUMMARY

[0004] An aspect of the present disclosure relates to an electronic component. The electronic component comprises a semiconductor substrate having a first major surface and a second major surface opposing the first major surface. The electronic component further comprises an electronic circuit integrated into the semiconductor substrate and a mirror arranged over the semiconductor substrate, wherein at least a portion of the mirror is inclined with respect to the first major surface of the semiconductor substrate. The electronic component further comprises at least one electrical contact configured to provide an electrical coupling between the electronic circuit and at least one electrode of an ion trap.

[0005] A further aspect of the present disclosure relates to an ion trap comprising the electronic component of the previous aspect.

[0006] A further aspect of the present disclosure relates to a device for controlling trapped ions. The device comprises a substrate having a main surface and a structured electrode layer arranged over the main surface of the substrate, wherein the structured electrode layer forms a plurality of electrodes of an ion trap configured to trap ions in a zone above the structured electrode layer. The device further comprises at least one electronic component arranged over the main surface of the substrate. Each of the at least one electronic component comprises a semiconductor substrate and an electronic circuit integrated into the semiconductor substrate, wherein the electronic circuit is electrically coupled to at least one of the plurality of electrodes of the ion trap. In addition, each of the at least one electronic component comprises a mirror arranged over the semiconductor substrate, wherein at least a portion of the mirror is inclined with respect to the main surface of the substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Devices in accordance with the disclosure are described in more detail below based on the drawings. Similar reference numerals may designate corresponding similar parts. The technical features of the various illustrated examples may be combined unless they exclude each other and/or can be selectively omitted if not described to be necessarily required.

[0008] FIG. 1 schematically illustrates a top view of a structured electrode layer of a device for controlling trapped ions in accordance with the disclosure.

[0009] FIGS. 2A and 2B schematically illustrate a top view (FIG. 2A) and a side view (FIG. 2B) of an electronic component in accordance with the disclosure.

[0010] FIGS. 3A and 3B schematically illustrate a top view (FIG. 3A) and a side view (FIG. 3B) of a device for controlling trapped ions in accordance with the disclosure.

[0011] FIGS. 4A and 4B schematically illustrate a top view (FIG. 4A) and a side view (FIG. 4B) of a device for controlling trapped ions in accordance with the disclosure.

[0012] FIGS. 5A and 5B schematically illustrate a top view (FIG. 5A) and a side view (FIG. 5B) of a device for controlling trapped ions in accordance with the disclosure.

[0013] FIGS. 6A and 6B schematically illustrate a top view (FIG. 6A) and a side view (FIG. 6B) of a device for controlling trapped ions in accordance with the disclosure.

[0014] FIGS. 7A and 7B schematically illustrate a top view (FIG. 7A) and a side view (FIG. 7B) of a device for controlling trapped ions in accordance with the disclosure.

[0015] FIGS. 8A and 8B schematically illustrate a top view (FIG. 8A) and a side view (FIG. 8B) of a device for controlling trapped ions in accordance with the disclosure.

[0016] FIGS. 9A to 9C schematically illustrate a top view (FIG. 9A) of a device for controlling trapped ions in accordance with the disclosure, as well as a top view (FIG. 9B) and a side view (FIG. 9C) of a detail of the device.

DETAILED DESCRIPTION

[0017] The following description relates to electronic components that may be included in devices for controlling trapped ions (ion trap devices). In addition, the following description relates to ion trap devices including such electronic components. The ion trap devices described herein may be configured to trap ions and control the trapped ions. It is to be noted that the following description is not restricted to ions, but may also be applied to atoms, molecules or other quantum particles/systems (e.g. electrons or defect centers). In particular, the trapped ions may be physically grouped in ion chains (or ion strings or ion crystals), wherein an ion chain may include one or more ions. In the following, the terms “ion” and “ion chain” may be used interchangeably at times.

[0018] In some examples, the devices described herein may be used for quantum computing, but are not restricted thereto. Trapped ions are one of the most promising candidates for being used as qubits in quantum computers, since they can be trapped with rather long lifetimes by means of electromagnetic fields. In this context, each ion may represent a physical qubit. However, ion trap devices in accordance with the disclosure are not restricted to the application of quantum computing. The ion trap devices presented herein may also be used for other applications, such as atomic clocks.

[0019] The ions trapped in an ion trap device may be shuttled (or transported) along shuttling paths of the device. The shuttling paths may include straight sequences, but also junctions, such as X-junctions, Y-junctions and/or T-junctions. For example, the shuttling paths may extend above a structured electrode layer including a plurality of electrodes. In particular, a shuttling path may be arranged in a plane over such structured electrode layer. Time-dependent electric fields may be used for shuttling ions along the shuttling paths. A shuttling of ions may be controlled by electric voltages applied to the electrodes of the structured electrode layer. In some examples, the ions may be moved along shuttling paths by means of AC and DC voltages that may be separately coupled to specific electrodes of the structured electrode layer. For example, the structured electrode layer may include RF electrodes for RF trapping and DC electrodes for static electric-field trapping and/or for moving the ions within the ion trap. As another example, ions may be confined by the combination of an external magnetic field and electrostatic quadrupole fields generated by voltages applied to DC electrodes. Ion trap devices as described herein may be configured to trap a plurality of ions that may be individually addressable and movable by appropriately controlling the electric potentials of the electrodes.

[0020] In a specific but non-limiting example, ion trap devices as described herein may correspond to or may include a surface ion trap (or surface-electrode ion trap). In surface ion traps, all

electrodes (i.e. the DC electrodes and the RF electrodes) may be arranged in a same single plane. The ions may be stored and shuttled above this single plane. However, it is to be understood that the concepts described herein are not restricted to surface ion traps. In further examples, devices for controlling trapped ions in accordance with the disclosure may also be based on three-dimensional ion trap geometries (e.g., where two or more trapping planes are arranged on top of each other).

[0021] Referring now to FIG. 1, a top view of an exemplary structured electrode layer of an ion trap device is illustrated. The structured electrode layer may include a first RF electrode **2A** and a second RF electrode **2B** extending along a first direction, for example in the x-direction. A shuttling path of the ion trap device may extend along the RF electrodes **2A** and **2B**. In addition, the structured electrode layer may include a center DC electrode **4** arranged between the first RF electrode **2A** and the second RF electrode **2B**. The center DC electrode **4** may extend along the x-direction in parallel to the RF electrodes **2A** and **2B**. In particular, the center DC electrode **4** may be arranged directly below ions trapped above the structured electrode layer. The structured electrode layer may further include a first plurality of first DC electrodes **6A** arranged adjacent to the first RF electrode **2A** and a second plurality of second DC electrodes **6B** arranged adjacent to the second RF electrode **2B**. The DC electrodes **6A** and **6B** may be referred to as DC shuttling electrodes and may be arranged opposite to the center DC electrode **4**. In practice, dozens, hundreds or even thousands of DC electrodes **6A** and **6B** may be arranged in an ion trap device along RF electrodes **2A** and **2B**.

[0022] In the illustrated example, the structured electrode layer may be referred to as five-wire geometry. It is however to be understood that aspects described herein are not restricted to five-wire geometries, but may also hold true for other types of ion traps and/or electrode geometries. In one case, the structured electrode layer may be fabricated with a single structured, highly conductive metal layer for low manufacturing complexity. In further cases, several structured metal layers may be required for more complex ion trap geometries. For example, the structured electrode layer may include or may be made of Al, AlCu, AlSiCu, W, TiW, Ti, TiN, Cu, Pt, Pd and/or Au. In a non-limiting example, dimensions of the illustrated structured electrode layer in the x- and y-direction may be about 300 μm and about 500 μm , respectively. However, these values are in no way limiting and are only intended to illustrate an order of magnitude of the dimensions of the illustrated arrangement.

[0023] Devices for controlling trapped ions may include different zones providing basic functionalities required for an operation of the ion trap device. In the illustrated example, such zones are exemplarily indicated by a solid rectangle and two dashed rectangles. The solid rectangle may e.g. indicate a processing zone **8** of the ion trap device. For example, the processing zone **8** may include or may correspond to a gate zone for performing quantum gate operations on ions trapped in the gate zone. Each of the dashed rectangles may indicate a shuttling and/or splitting zone **10** in which an ion chain may be shuttled, split and/or recombined. It is to be understood that ion trap devices as described herein may also include other types of zones, such as memory zones, state readout zones, loading zones, etc. It is further to be understood that the illustrated example of FIG. 1 only shows a basic portion of an ion trap device. In practice, ion trap devices may include a plurality of shuttling paths which may be connected by junctions, such as T-junctions, Y-junctions and/or X-junctions.

[0024] FIGS. 2A and 2B illustrate a top view and a side view, respectively, of an electronic component **200** in accordance with the disclosure. In particular, the electronic component **200** may be included in a device for controlling trapped ions. The electronic component **200** may include a semiconductor substrate **12** having a first major surface **14A** and a second major surface **14B** opposing the first major surface **14A**. For example, the semiconductor substrate **12** may include or may be made of silicon. The electronic component **200** may further include an electronic circuit **16** integrated into the semiconductor substrate **12** and a mirror **18** arranged over the semiconductor substrate **12**. At least a portion **18i** of the mirror **18** may be inclined with respect to the first major surface **14A** of the semiconductor substrate **12**. Furthermore, the electronic component **200** may

include at least one electrical contact **20** configured to provide an electrical coupling between the electronic circuit **16** and at least one electrode of an ion trap.

[0025] The electronic circuit **16** may be configured to provide one or multiple electronic functions. For example, the electronic circuit **16** may include a multiplexer configured to provide a multiplexing between a plurality of DC electrodes of an ion trap and one or multiple voltage sources. In this regard, the electronic circuit **16** (or the multiplexer included therein) may include a digital interface. Referring back to the example of FIG. **1**, one or more voltage sources may provide DC voltages that need to be distributed to the multiple DC electrodes **6A** and **6B**. The multiplexing feature of the electronic circuit **16** may provide such distribution of voltage signals. As will become apparent later on, the large number of hundreds or even thousands of DC electrodes may be broken down to only a few or dozens of signal lines and/or contacts coupled to the voltage sources.

[0026] The electronic circuit **16** is not necessarily restricted to a pure distribution of DC voltages to the DC electrodes **6A** and **6B**. The electronic circuit **16** may further include digital electronics for controlling the one or multiple voltage sources. For example, control logic of the electronic circuit **16** may be configured to provide stored wave forms that may be used for a suitable generation of DC voltages. In some examples, the electronic circuit **16** may include a plurality of digital-to-analog converters configured to generate DC voltages that may be provided to the DC electrodes **6A** and **6B**.

[0027] The electronic circuit **16** may include or may correspond to an integrated circuit. The electronic circuit **16** may be integrated into the first major surface **14A** of the semiconductor substrate **12** as indicated by a dot-dashed line in the exemplary side view of FIG. **2B**. In the top view of FIG. **2A**, the electronic circuit **16** is exemplarily illustrated as a rectangular integrated circuit having electronic contacts on opposing sides. However, it is to be understood that the chosen illustration only serves as a simplified representation of the electronic circuit **16** and is in no way limiting.

[0028] The mirror **18** may include or may correspond to at least one reflective surface. The mirror **18** may be manufactured from any suitable reflective material, such as a metal. The mirror **18** may be configured to reflect incoming laser light hitting the reflective surface of the mirror **18**. In this regard, the mirror **18** may be configured to reflect at least 97% or at least 98% or at least 99% or at least 99.5% percent of incident laser light. In some examples, the mirror **18** may be configured to reflect substantially 100% of the incident laser light. The laser light may have a wavelength in a range from about 150 nm to about 4 μm , but is in general not restricted to a specific value or range of values. For example, the laser light may be used for processing and/or cooling ions trapped above a structured electrode layer.

[0029] In the illustrated example, one or more inclined portions **18i** of the mirror **18** may include a flat surface and may be arranged directly on the semiconductor substrate **12**. For example, the semiconductor substrate **12** may be etched by a wet etching and/or a chemical etching technique such that an inclined surface of the substrate **12** may be provided onto which the mirror **18** may be applied. In other words, the backside of substrate **12** (i.e., the side opposite to the side where electronic circuit **16** is formed) of electronic component **200** may be processed so that an inclined portion may be formed. On this inclined portion of substrate **12** a mirror **18** with inclined portion **18i** can be fabricated. Additionally, or alternatively, one or more inclined portions **18i** of the mirror **18** may include a concave surface. In order to realize such concave surface, a glass substrate (not illustrated) may be arranged between the semiconductor substrate **12** and the mirror **18**. The glass substrate may include a concave surface onto which the mirror **18** may be deposited.

[0030] In the illustrated example, an angle θ between the flat surface of the inclined mirror portion **18i** and the first major surface **14A** of the semiconductor substrate **12** may have a non-limiting value of about 45 degrees. In further examples, the angle θ may differ and may generally have a value in a range from about 20 degrees to about 160 degrees. The angle θ may be chosen depending on the direction of incoming laser light that is to be reflected by the mirror **18**. In one case, laser

light may hit the mirror **18** in the negative z-direction (i.e. from top to bottom) and the angle θ may be smaller than 90 degrees. In another case, laser light may hit the mirror **18** in the positive z-direction (i.e. from bottom to top) and the angle θ may be greater than 90 degrees.

[0031] The inclined portion **18i** of the mirror **18** may cover a side surface of the semiconductor substrate **12** extending between the first major surface **14A** and the second major surface **14B**. In the illustrated example, the mirror **18** may cover three side surfaces and the top surface of the semiconductor substrate **12**. In further examples, the top surface of the semiconductor substrate **12** may remain uncovered by the mirror **18**. In yet further examples, only one or more selected side surfaces of the semiconductor substrate **12** may be covered by the mirror **18** depending on the desired directions of laser light reflection.

[0032] In the exemplary top view of FIG. 2A, the electronic component **200** may have the shape of a square or a rectangle. In a non-limiting example, dimensions of the electronic component **200** in the x- and y-direction may be in a range from about 500 μm to about 1 mm each. However, these values are in no way limiting and are only intended to illustrate an order of magnitude of the dimensions of the illustrated arrangement. In further examples, the shape and dimensions of the electronic component **200** may differ depending on the considered application.

[0033] Referring now to FIGS. 3A to 5B, various examples for connecting and interfacing an electronic component in accordance with the disclosure to a structured electronic layer of an ion trap device are illustrated and described. It is to be understood that the described examples may be combined to further examples, provided this is not technically or logically excluded.

[0034] FIGS. 3A and 3B illustrate a device for controlling trapped ions in accordance with the disclosure. The device **300** may include a substrate **22** having a main surface **24**. The substrate **22** may include or may be made of a dielectric material or a semiconductor material. For example, the substrate **22** may include or may be made of at least one of silicon, silicon carbide, fused silica, sapphire, glass, aluminum nitride, diamond, or the like. A structured electrode layer may be arranged over the main surface **24** of the substrate **22**. The structured electrode layer may form a plurality of electrodes of an ion trap configured to trap ions in a zone above the structured electrode layer. For example, the structured electrode layer may be similar to the example of FIG. 1, but is not restricted thereto. In the illustrated example, one RF electrode **2** and a plurality of DC electrodes **6** are shown, while further possible RF and DC electrodes are not illustrated for the sake of simplicity.

[0035] The device **300** may include at least one electronic component **30** which may be at least partially similar to the example of FIGS. 2A-2B and may include the same or similar components. The electronic component **30** may be arranged over the main surface **24** of the substrate **22**, wherein the first major surface **14A** of the electronic component **30** including the integrated electronic circuit **16** may face towards the main surface **24**. In the illustrated example, the electrical contacts **20** of the electronic component **30** may include or may correspond to flip chip contact pads and the electronic component **30** may be mounted on the main surface **24** of the substrate **22** by a flip chip technique. The electrical contacts **20** may be configured to provide an electrical coupling between the electronic circuit **16** and the structured electrode layer. In particular, the electrical contacts **20** may be electrically coupled to the DC electrodes **6** of the structured electrode layer.

[0036] The device **300** may further include a plurality of electrical contacts **26** that may be electrically coupled to the electronic circuit **16**. In the top view of FIG. 3A, the electronic circuit **16** is illustrated as a symbol only for the sake of simplicity. However, it is to be noted that the electrical contacts **26** and also the DC electrodes **6** may be electrically connected to the logic area of the electronic circuit **16** even though details of such electrical connections are not explicitly illustrated. In one example, such electrical connections may be provided by an electrical redistribution layer that may be arranged at or on the first major surface **14A** and may be configured to electrically connect the logic area of the electronic circuit **16** to the DC electrodes **6**

and/or the electrical contacts **26**. In the illustrated example, the electrical contacts **26** may include or may correspond to wire bond pads that may be formed in a same metal layer as the structured electrode layer of the ion trap. In the top view of FIG. 3A, the electrical contacts **26** may be arranged at or adjacent to the outline of the electronic component **30**. In the illustrated example, an exemplary number of five electrical contacts **26** may be arranged at the left and the right side of the electronic component **30**, respectively. In further examples, the number of electrical contacts **26** may differ. The electrical contacts **26** may be configured to provide an electrical access to the electronic circuit **16** from outside of the electronic component **30**. For example, one or more voltage sources may be electrically coupled to the electrical contacts **26**.

[0037] A multiplexer of the electronic circuit **16** may provide a multiplexing between the one or more voltage sources connected to the electrical contacts **26** and the plurality of DC electrodes **6** connected to the electrical contacts **20**. Accordingly, DC voltages required for a proper shuttling of trapped ions along shuttling paths may be distributed to the DC electrodes **6**. In practice, the number of DC electrodes **6** may be higher than the number of electrical contacts **26** by one or more orders of magnitude. For example, the structured electrode layer may include hundreds or even thousands of DC electrodes, while the number of electrical contacts **26** may be in the range of only a few or dozens. For example, a multiplexer of electronic circuit **16** may provide a same voltage or a same voltage profile to a plurality of DC electrodes **6**. The high number of DC electrodes **6** may thus become accessible via the smaller number of electrical contacts **26**. The comparatively smaller number of electrical contacts **26** may support a scaling of the ion trap device to larger sizes.

[0038] The mirror **18** of the electronic component **30** may be configured to receive laser light in a direction substantially non-parallel to the main surface **24** of the substrate **22** and to redirect the laser light in a direction substantially parallel to the main surface **24**. In the illustrated example, laser light hitting the reflective surface of the mirror **18** in the (negative) z-direction may be reflected and redirected in the (negative) y-direction. Additionally, or alternatively, the mirror **18** of the electronic component **30** may be configured to receive laser light in a direction substantially parallel to the main surface **24** of the substrate **22** and to redirect the laser light in a direction substantially non-parallel to the main surface **24**. In the illustrated example, laser light hitting the reflective surface of the mirror **18** in the (positive) y-direction may be reflected and redirected in the (positive) z-direction. It is to be understood that incident laser light may be reflected by the mirror **18** in various other ways depending on the angle θ between the inclined portion **18i** of the mirror **18** and the first major surface **14A** of the electronic component **30**. Possible values for the angle θ have been previously discussed in connection with the example of FIGS. 2A-2B.

[0039] In the exemplary side view of FIG. 3B, the main surface **24** of the substrate **22** may be substantially planar. In further examples, the substrate **22** may include one or more recesses formed in the main surface **24**. The electronic component **30** may be arranged in such recess, wherein a distance between the main surface **24** and a laser beam reflected by the mirror **18** and extending parallel to the main surface **24** may be reduced. In this way, the laser light may reach ions trapped over the structured electrode layer at lower heights. For example, ions may be trapped over the main surface **24** of the substrate **22** at heights of about 100 μm or smaller.

[0040] The electronic component **30** used in the ion trap device **300** may provide a combination of two technical functions. First, the electronic component **30** may provide an efficient and simplified electrical access to the high number of DC electrodes **6** via only a few electrical contacts **26**. No (or only few) additional electrical connections, such as Through Silicon Vias (TSVs) and/or Through Glass Vias (TGVs), may thus be necessary for a suitable electrical access of the DC electrodes **6**. Second, the electronic component **30** may provide an efficient and simplified way for laser light to access the ion trap device and reach the ions trapped over the structured electrode layer. Incoming laser light may be directed in various ways by choosing a suitable orientation of the inclined portion of the mirror **18**. Accordingly, a number of photonic layers integrated into the ion trap device may be reduced. In particular, an effort of fabricating a high number of gratings and

waveguides and connecting them with optical fibers may be reduced. It is to be noted that the described electrical and optical access combined in only one component may support a scaling of ion traps to larger sizes.

[0041] The device **300** may be manufactured in various ways. In a first example of production, a first wafer including a plurality of electronic components **30** may be wafer-bonded to a second wafer including the substrate **22** and a plurality of structured electronic layers arranged thereon. After wafer-bonding, the wafer stack may be singulated (e.g. diced) into a plurality of ion trap devices as e.g. shown in the example of FIGS. **3A-3B**. It is to be understood that a bonding of the wafers is not restricted to a specific technique and may naturally depend on the materials of the involved wafers. In general, any suitable wafer-bonding technique may be used, such as one or more of glass-bonding techniques, eutectic bonding techniques, anodic bonding techniques, thermocompression bonding techniques, or the like. In one example, anodic bonding may be performed for forming an anodic aluminum/glass wafer-bond. In a further example, a different wafer-bonding technique may be used for enabling sapphire substrates providing superior thermal conductivity. In a second example of production, each of the first wafer and the second wafer may be diced into a plurality of electronic components and structured electronic layers on a substrate, respectively. Afterwards, the singulated elements may be chip-stacked to form arrangements as e.g. illustrated in the example of FIGS. **3A-3B**. In this regard, it is to be noted that the first example of wafer-bonding may provide an improved accuracy in terms of alignment and tilt.

[0042] FIGS. **4A** and **4B** schematically illustrate a top view and a side view, respectively, of a device **400** for controlling trapped ions in accordance with the disclosure. The device **400** may include some or all features of previously described devices. An electronic component **40** of the device **400** may include one or more electrical contacts **28** arranged over the second major surface **14B** of the semiconductor substrate **12**. For example, the electrical contacts **28** may include or may correspond to wire bond pads. Furthermore, the electronic component **40** may include one or more electrical connection elements **32** extending through the semiconductor substrate **12**. For example, the electrical connection elements **32** may include or may correspond to TSVs. The electrical connection elements **32** may provide an electrical coupling between the electronic circuit **16** and the electrical contacts **28** arranged at the second major surface **14B**. Accordingly, at least some of the DC electrodes **6** may be electrically accessible via the electrical contacts **28** at the second major surface **14B**. In particular, one or more voltage sources may be electrically connected to the wire bond pads **28** via wires.

[0043] FIGS. **5A** and **5B** schematically illustrate a top view and a side view, respectively, of a device **500** for controlling trapped ions in accordance with the disclosure. The device **500** may include some or all features of previously described devices. The device **500** may include one or more electrical connection elements **34** extending through the substrate **22**. For example, the electrical connection elements **34** may include or may correspond to TSVs and/or TGVs depending on the material of the substrate **22**. In particular, the electrical connection elements **34** may extend from the upper surface of the substrate **22** to its bottom surface, thereby providing an electrical connection between the electrical contacts **20** of the electronic component **50** and electrical contacts **36** arranged at the bottom surface of the substrate **22**. In one example, the electrical contacts **36** may include or may correspond to flip chip contacts pads onto which one or multiple solder balls may be placed.

[0044] Referring now to FIGS. **6A** to **9C**, various applications are illustrated in which electronic components in accordance with the disclosure may be used. The illustrated examples relate to different types of ion traps suitable for an application in a quantum processing unit or an atomic clock.

[0045] FIGS. **6A** and **6B** illustrate a top view and a side view, respectively, of a device **600** for controlling trapped ions in accordance with the disclosure. The device **600** may include a substrate **22** and a structured electrode layer arranged over the upper main surface of the substrate **22**. In one

example, the structured electrode layer of FIGS. 6A-6B may be based on a five-wire geometry as previously described in connection with FIGS. 2A-2B. The plurality of electrodes may be part of an ion trap configured to trap ions **38** in a zone above the structured electrode layer. The trapped ions **38** may be shuttled along the x-direction by means of suitable AC and DC voltages separately applied to the RF and DC electrodes of the structured electrode layer, respectively. The displayed trap design may avoid RF/DC routing crossings in order to facilitate low RF capacitance and/or low RF/DC pickup. The device **600** may include one or multiple processing zones **8** arranged along the x-direction. The device **600** may be referred to as one-dimensional ion trap (in particular **1D** Top-IC Trap) and may be used for quantum computing in one example.

[0046] In the illustrated case, the structured electrode layer may extend along the x-direction. The device **600** may include two electronic components **60A** and **60B** that may be arranged on opposite sides of the structured electrode layer. In particular, the electronic components **60A** and **60B** may be designed and arranged symmetrically with respect to a shuttling path extending along the x-direction above the center DC electrode **6**. In the exemplary top view of FIG. 6A, each of the electronic components **60A** and **60B** may have a rectangular footprint. The inclined portions **18i** of the mirrors **18A** and **18B** may extend over three side surfaces of the respective semiconductor substrate **12**. Incoming laser beams may reach the trapped ions **38** by being reflected at the tilted sidewalls of the mirrors **18A** and **18B**. In addition, some axial laser access in the x-direction may be maintained. The electronic components **60A** and **60B** may have been obtained from dicing a wafer stack such that a fourth side surface of the respective semiconductor substrate **12** may include a dicing edge **42**.

[0047] In a non-limiting example, dimensions of the device **600** and the electronic components **60A**, **60B** may be as follows. Dimensions *l* and *w* of the device **600** in the x- and y-direction may be about 4.5 mm and 1.5 mm, respectively. A dimension *h.sub.1* of the substrate **22** in the z-direction may be greater than about 400 μm . A dimension *h.sub.2* of the electronic components **60A** and **60B** in the z-direction may be about 200 μm . However, these values are in no way limiting and are only intended to illustrate an order of magnitude of the dimensions of the illustrated arrangement.

[0048] FIGS. 7A and 7B illustrate a top view and a side view, respectively, of a device **700** for controlling trapped ions in accordance with the disclosure. The device **700** may include a substrate **22** and a structured electrode layer arranged over the substrate **22**. In addition, the device **700** may include a plurality of electronic components **70** which may be arranged over the structured electrode layer in a rectangular grid pattern. In the illustrated example, each electronic component **70** may have a rectangular footprint and may be at least partially similar to electronic components described in connection with previous examples.

[0049] In the illustrated example, ions **38** trapped above the structured electrode layer may be shuttled along shuttling paths extending in the x- and y-direction. In particular, the structured electrode layer may form a rectangular grid pattern of shuttling paths extending between the plurality of electronic components **70**. A plurality of X-junctions may be arranged at intersections of the shuttling paths. The illustrated ion trap layout using X-junctions may provide shorter ion-shuttling distances and fewer SWAP operations compared to other ion trap layouts. The device **700** may further include a plurality of processing zones **8** that may be arranged between opposite electronic components **70**. The device **700** may be referred to as 3X Top-IC Trap and may e.g. be used for quantum computing.

[0050] Similar to previous examples, the trap design of FIGS. 7A-7B may be free of RF/DC crossings and may show low RF capacitance. A production of the device **700** may be based on a single-layer trap fabrication. In a non-limiting example, dimensions of the device **700** and the electronic components **70** may be as follows. Dimensions *l* and *w* of the device **700** in the x- and y-direction may be about 3.5 mm and 1.5 mm, respectively. A dimension *h.sub.1* of the substrate **22** in the z-direction may be greater than about 400 μm . A dimension *h.sub.2* of the electronic

components **70** in the z-direction may be about 100 μm . However, these values are in no way limiting and are only intended to illustrate an order of magnitude of the dimensions of the illustrated arrangement.

[0051] FIGS. **8A** and **8B** illustrate a top view and a side view, respectively, of a device **800** for controlling trapped ions in accordance with the disclosure. The device **800** may include a first substrate **22A** and a first structured electrode layer arranged over the first substrate **22A**. In addition, the device **800** may include two electronic components **80A** and **80B** arranged over the first structured electrode layer. The first structured electrode layer and the electronic components **80A**, **80B** of FIGS. **8A-8B** may be at least partially similar to corresponding components described in connection with the example of FIGS. **6A-6B**.

[0052] The device **800** may further include a second substrate **22B** arranged over the electronic components **80A** and **80B**. The second substrate **22B** may be at least partially similar to the first substrate **22A**. The electronic components **80A** and **80B** may serve as spacer elements configured to define a space **44** between the two substrates **22A** and **22B** in which ions **38** may be trapped. In one example, the substrates **22A** and **22B** may be made of a glass material such that the illustrated arrangement may correspond to a glass-silicon-glass stack providing low RF capacitance and low CTE warpage. A second structured electrode layer may be arranged on a bottom surface of the second substrate **22B**. In one example, the second structured electrode layer may be limited to RF electrodes and few ground electrodes, but may not necessarily include DC electrodes. The device **800** may be referred to as three-dimensional ion trap and, for example, may be used for an atomic clock application.

[0053] Each of the substrates **22A** and **22B** may include one or more openings **46A** and **46B** configured to provide optical access to the interior space **44**. Laser light may pass through the openings **46A** and **46B** and reach the trapped ions **38**. In the illustrated example, laser light passing through the central openings **46A** of the substrates **22A** and **22B** may directly hit the ions **38** in a substantially vertical direction. Alternatively, or additionally, laser light may pass through the laterally displaced openings **46B** of the second substrate **22B** and may be redirected by the mirrors **18** of the electronic components **80A** and **80B** to hit the ions **38** in a substantially horizontal direction. For example, the top and bottom optical access through openings **46A** and **46B** may allow loading of ions and detection of the states of the ions **38**.

[0054] In a non-limiting example, dimensions of the device **800** and the electronic components **80A**, **80B** may be as follows. Dimensions *l* and *w* of the device **800** in the x- and y-direction may be about 4.5 mm and 1.5 mm, respectively. A dimension *h.sub.1* of the first substrate **22A** in the z-direction may be about 400 μm . A dimension *h.sub.2* of the electronic components **80A** and **80B** in the z-direction may be about 400 μm . A dimension *h.sub.3* of the second substrate **22B** in the z-direction may be about 400 μm . However, these values are in no way limiting and are only intended to illustrate an order of magnitude of the dimensions of the illustrated arrangement.

[0055] FIG. **9A** schematically illustrates a top view **900** of a device for controlling trapped ions in accordance with the disclosure. The device **900** may include a plurality of electronic components **90** which may be arranged over a structured electrode layer. The structured electrode layer may form a rectangular grid pattern of shuttling paths extending between the electronic components **90**. A plurality of X-junctions may be arranged at intersections of the shuttling paths. It is to be noted that the example of FIG. **9A** shows an efficient and simple scaling of an ion trap device to larger sizes using a plurality of electronic components **90** in accordance with the disclosure. The ion trap device may thus be referred to as large scale ion trap.

[0056] FIGS. **9B** and **9C** schematically illustrates a top view and a side view, respectively, of a detail of the ion trap device **900**. More particular, a structured electrode layer including RF electrodes **2** and DC electrodes **6** as well as an electronic component **90** arranged thereon are shown. In the exemplary top view of FIG. **9B**, the electronic component **90** may have a rectangular footprint and the inclined portion **18i** of the mirror **18** may extend over all four side surfaces of the

semiconductor substrate **12**. In the illustrated example, the structured electrode layer and/or the electronic component **90** may be electrically accessible via electrical connection elements **34** (such as TSVs and/or TGVs) extending through the substrate **22** and electrical contacts **36** arranged on the bottom surface of the substrate **22**. The illustrated arrangement may include further elements which have been described in connection with previous examples.

[0057] In a non-limiting example, dimensions of the device **900** and the electronic components **90** may be as follows. Dimensions l.sub.1 and w.sub.1 of the device **900** in the x- and y-direction may be in a range from about 5 mm to about 10 mm each. Dimensions l.sub.2 and w.sub.2 of the electronic component **90** in the x- and y-direction may be in a range from about 1 mm to about 2 mm each. A dimension h.sub.1 of the substrate **22** in the z-direction may be greater than about 400 μm . A dimension h.sub.2 of the electronic components **90** in the z-direction may be about 100 μm . However, these values are in no way limiting and are only intended to illustrate an order of magnitude of the dimensions of the illustrated arrangement.

[0058] It is to be noted that one or more components of the devices described herein may be at least partially manufactured based on a micro-fabrication technique. For example, structured electrode layers as previously described may be formed based on a micro-fabrication technique. Micro-fabrication techniques (e.g. for electrode formation and structuring) may include one or more of photolithography processes (such as photoresist application, patterning, etching), deposition processes (such as CVD, PVD, sputtering), plating processes (such as electroless plating, galvanic plating), etching processes, or the like. Electrodes formed in a structured electrode layer may be structured with micrometer or sub-micrometer scale precision and alignment accuracy, thereby allowing complex electrode layouts without loss of controllability of trapped ions.

[0059] It is to be understood that devices in accordance with the disclosure may include various additional components which may have not been illustrated and described in detail for the sake of simplicity. For example, the described devices may further include at least one of a memory zone, a gate zone, a state readout zone, a loading zone, a laser source, a microwave source, means for controlling voltages applied to DC electrodes and/or RF electrodes, X-junctions, T-junctions, etc.

[0060] In the following, electronic components, ion traps including such electronic components and devices for controlling trapped ions are described by means of examples.

[0061] Example 1 is an electronic component, comprising: a semiconductor substrate having a first major surface and a second major surface opposing the first major surface; an electronic circuit integrated into the semiconductor substrate; and a mirror arranged over the semiconductor substrate, wherein at least a portion of the mirror is inclined with respect to the first major surface of the semiconductor substrate; and at least one electrical contact configured to provide an electrical coupling between the electronic circuit and at least one electrode of an ion trap.

[0062] Example 2 is an electronic component according to Example 1, wherein the electronic circuit comprises: a multiplexer configured to provide a multiplexing between a plurality of DC electrodes of the ion trap and one or more voltage sources.

[0063] Example 3 is an electronic component according to Example 2, wherein the electronic circuit further comprises at least one of: digital electronics for controlling the one or more voltage sources, a digital interface, or a plurality of digital-to-analog converters.

[0064] Example 4 is an electronic component according to one of the preceding Examples, wherein the electronic circuit is integrated into the first major surface of the semiconductor substrate.

[0065] Example 5 is an electronic component according to one of the preceding Examples, wherein the inclined portion of the mirror comprises a flat surface and is arranged directly on the semiconductor substrate.

[0066] Example 6 is an electronic component according to Example 5, wherein an angle between the flat surface of the inclined portion of the mirror and the first major surface of the semiconductor substrate is in a range from 20 degrees to 160 degrees.

[0067] Example 7 is an electronic component according to one of the preceding Examples, wherein

the inclined portion of the mirror covers a side surface of the semiconductor substrate extending between the first major surface and the second major surface.

[0068] Example 8 is an electronic component according to one of the preceding Examples, wherein: a footprint of the electronic component is rectangular, the inclined portion of the mirror extends over three side surfaces of the semiconductor substrate, and a fourth side surface of the semiconductor substrate comprises a dicing edge.

[0069] Example 9 is an electronic component according to one of Examples 1 to 7, wherein: a footprint of the electronic component is rectangular, and the inclined portion of the mirror extends over all four side surfaces of the semiconductor substrate.

[0070] Example 10 is an electronic component according to one of the preceding Examples, further comprising: at least one electrical contact arranged over the second major surface of the semiconductor substrate; and at least one electrical connection element extending through the semiconductor substrate and providing an electrical coupling between the electronic circuit and the at least one electrical contact.

[0071] Example 11 is an ion trap, comprising the electronic component of one of the preceding Examples.

[0072] Example 12 is a device for controlling trapped ions, the device comprising: a substrate having a main surface; a structured electrode layer arranged over the main surface of the substrate, wherein the structured electrode layer forms a plurality of electrodes of an ion trap configured to trap ions in a zone above the structured electrode layer; at least one electronic component arranged over the main surface of the substrate, wherein each of the at least one electronic component comprises: a semiconductor substrate; an electronic circuit integrated into the semiconductor substrate, wherein the electronic circuit is electrically coupled to at least one of the plurality of electrodes of the ion trap; and a mirror arranged over the semiconductor substrate, wherein at least a portion of the mirror is inclined with respect to the main surface of the substrate.

[0073] Example 13 is a device according to Example 12, wherein the mirror is configured to: receive laser light in a direction substantially non-parallel to the main surface of the substrate and to redirect the laser light in a direction substantially parallel to the main surface of the substrate, and/or receive laser light in a direction substantially parallel to the main surface of the substrate and to redirect the laser light in a direction substantially non-parallel to the main surface of the substrate.

[0074] Example 14 is a device according to Example 12 or 13, wherein the substrate comprises a recess and the at least one electronic component is arranged in the recess.

[0075] Example 15 is a device according to one of Examples 12 to 14, wherein: the structured electrode layer extends along one direction, and the at least one electronic component comprises two electronic components arranged on opposite sides of the structured electrode layer.

[0076] Example 16 is a device according to one of Examples 12 to 14, wherein: the at least one electronic component comprises a plurality of electronic components arranged in a rectangular grid pattern, and the structured electrode layer forms a rectangular grid pattern of shuttling paths extending between the plurality of electronic components.

[0077] As used herein, the terms “having”, “containing”, “including”, “comprising” and the like are open ended terms that indicate the presence of stated elements or features, but do not preclude additional elements or features. The articles “a”, “an” and “the” are intended to include the plural as well as the singular, unless the context clearly indicates otherwise.

[0078] The expression “and/or” should be interpreted to include all possible conjunctive and disjunctive combinations, unless expressly noted otherwise. For example, the expression “A and/or B” should be interpreted to mean only A, only B, or both A and B. The expression “at least one of” should be interpreted in the same manner as “and/or”, unless expressly noted otherwise. For example, the expression “at least one of A and B” should be interpreted to mean only A, only B, or both A and B.

[0079] The features of the various examples described herein may be combined with each other unless specifically noted otherwise. Although specific examples have been illustrated and described herein, it will be appreciated by those of ordinary skill in the art that a variety of alternate and/or equivalent implementations may be substituted for the specific examples shown and described without departing from the scope of the present invention. This application is intended to cover any adaptations or variations of the specific examples discussed herein. Therefore, it is intended that this invention be limited only by the claims and the equivalents thereof.

Claims

1. An electronic component, comprising: a semiconductor substrate having a first major surface and a second major surface opposing the first major surface; an electronic circuit integrated into the semiconductor substrate; and a mirror arranged over the semiconductor substrate, wherein at least a portion of the mirror is inclined with respect to the first major surface of the semiconductor substrate; and at least one electrical contact configured to provide an electrical coupling between the electronic circuit and at least one electrode of an ion trap.
2. The electronic component of claim 1, wherein the electronic circuit comprises: a multiplexer configured to provide a multiplexing between a plurality of DC electrodes of the ion trap and one or more voltage sources.
3. The electronic component of claim 2, wherein the electronic circuit further comprises: digital electronics configured to control the one or more voltage sources.
4. The electronic component of claim 2, wherein the electronic circuit further comprises: a digital interface.
5. The electronic component of claim 2, wherein the electronic circuit further comprises: a plurality of digital-to-analog converters.
6. The electronic component of claim 1, wherein the electronic circuit is integrated into the first major surface of the semiconductor substrate.
7. The electronic component of claim 1, wherein the inclined portion of the mirror comprises a flat surface and is arranged directly on the semiconductor substrate.
8. The electronic component of claim 7, wherein an angle between the flat surface of the inclined portion of the mirror and the first major surface of the semiconductor substrate is in a range of 20 degrees to 160 degrees.
9. The electronic component of claim 1, wherein the inclined portion of the mirror covers a side surface of the semiconductor substrate extending between the first major surface and the second major surface.
10. The electronic component of claim 1, wherein: a footprint of the electronic component is rectangular; the inclined portion of the mirror extends over three side surfaces of the semiconductor substrate; and a fourth side surface of the semiconductor substrate comprises a dicing edge.
11. The electronic component of claim 1, wherein: a footprint of the electronic component is rectangular; and the inclined portion of the mirror extends over all four side surfaces of the semiconductor substrate.
12. The electronic component of claim 1, further comprising: at least one electrical contact arranged over the second major surface of the semiconductor substrate; and at least one electrical connection element extending through the semiconductor substrate and providing an electrical coupling between the electronic circuit and the at least one electrical contact.
13. An ion trap, comprising the electronic component of claim 1.
14. A device for controlling trapped ions, the device comprising: a substrate having a main surface; a structured electrode layer arranged over the main surface of the substrate, wherein the structured electrode layer forms a plurality of electrodes of an ion trap configured to trap ions in a zone above the structured electrode layer; at least one electronic component arranged over the main surface of

the substrate, wherein each of the at least one electronic component comprises: a semiconductor substrate; an electronic circuit integrated into the semiconductor substrate, wherein the electronic circuit is electrically coupled to at least one of the plurality of electrodes of the ion trap; and a mirror arranged over the semiconductor substrate, wherein at least a portion of the mirror is inclined with respect to the main surface of the substrate.

15. The device of claim 14, wherein the mirror is configured to: receive laser light in a direction substantially non-parallel to the main surface of the substrate and to redirect the laser light in a direction substantially parallel to the main surface of the substrate.

16. The device of claim 14, wherein the mirror is configured to: receive laser light in a direction substantially parallel to the main surface of the substrate and to redirect the laser light in a direction substantially non-parallel to the main surface of the substrate.

17. The device of claim 14, wherein the substrate comprises a recess and the at least one electronic component is arranged in the recess.

18. The device of claim 14, wherein: the structured electrode layer extends along one direction; and the at least one electronic component comprises two electronic components arranged on opposite sides of the structured electrode layer.

19. The device of claim 14, wherein: the at least one electronic component comprises a plurality of electronic components arranged in a rectangular grid pattern; and the structured electrode layer forms a rectangular grid pattern of shuttling paths extending between the plurality of electronic components.

20. The device of claim 19, further comprising: a plurality of X-junctions arranged at intersections of the shuttling paths.
